Logic - a quick review

Note that there are many different notations for the basic logic operations. We will only list the most common.
There are many different kinds of logic. However, we typically focus on two:

- Propositional logic, or Boolean algebra - This includes Boolean constants, Boolean operations, plus rules of equivalences between formulas.
- Predicate logic, or first-order calculus - This include propositional logic, plus relations and quantifiers.

Boolean constants

- True - abbreviated as T, or sometimes written as 1
- False - abbreviated as F, or sometimes written as 0

Truth tables & Boolean operations

There are variations on the format for truth tables, but they map all combinations of the possible input(s) values to the resulting value for one or more formulas.

Unary negation:

p	$\bar{p}$, $\neg p$
F	T
T	F

Binary conjunction, disjunction, exclusive disjunction, implication and equivalence (respectively):

p	q	$p \wedge q$, pq , $p \cdot q$, *p AND q*	$p \vee q$, $p + q$, *p OR q*	$p \oplus q$, *p XOR q*	$p \rightarrow q$, *p implies q* , *if p then q*	$p \leftrightarrow q$, *p if and only if q* , *p iff q*
F	F	F	F	F	T	T
F	T	F	T	T	T	F
T	F	F	T	T	F	F
T	T	T	T	F	T	T

Relations

See the handout on relations and functions.

Quantifiers

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Some equivalences

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Proofs

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Soundness and completeness